重复题目全文清单（二次检查修订版）

以下每组重复题目均经"全文 + 配图 + 人工目检"三重确认；每组展示 1 份原文（含答案）。

一、AP Biology 库内重复 —— MCQ 38 组

1. 【U1 Chemistry of Life】U1 Q21 ＝ U8 Q26

21. Which of the following communities is likely to be most stable?

(A) One where a keystone predator has been removed

(B) One with uniformly spaced vegetation

(C) One with high species diversity ✓

(D) One that lacks decomposers

2. 【U1 Chemistry of Life】U1 Q41 ＝ U2 Q94

41. The fluid mosaic model of cell membranes describes proteins and steroids associated with a phospholipid bilayer. Which of the following membrane components affect membrane stability?

(A) Embedded proteins give plasma membranes less stability.

(B) Steroids, such as cholesterol, give membranes more stability. ✓

(C) Proteins in the hydrophilic region of the plasma membrane create more stability.

(D) Polar heads of phospholipids give plasma membranes less stability.

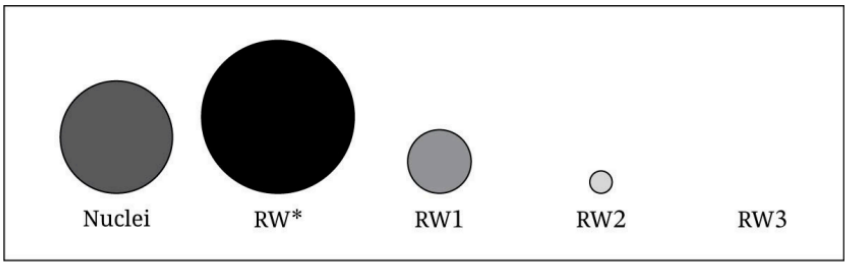
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3. 【U1 Chemistry of Life】U1 Q59 ＝ U8 Q90

59. A researcher conducted an experiment in a brightly lit aquarium containing a large population of single-celled green algae. Water containing radioactive phosphorus was added to the aquarium. After 7 days, the researcher took a water sample containing the algae and separated the algae from the radioactive water (RW*).

The researcher added deionized water to rinse the algae and saved the rinse water and then repeated that process twice. The rinse water from each of the three rinses was saved as RW1, RW2, and RW3, respectively. The researcher also removed and kept the nuclei of the algal cells. The nuclei were mixed with deionized water, and a drop of the mixture was placed on a piece of filter paper. A drop of the radioactive water and a drop of each of the rinse water samples were also placed on the same paper. These samples were exposed to X-ray to determine their radioactivity. The results after 24 hours are shown in Figure 1.

Figure 1. Radioactivity of saved algal water samples. Darker color and larger circle size indicate a higher level of radioactivity.



Based on Figure 1, which of the following statements best explains why the nuclei sample had a higher level of radioactivity than any of the rinse water samples?

(A) Phosphorus is used in the building of nucleic acids in the nucleus. ✓

(B) The nuclei sample was contaminated with radioactive water.

(C) Radioactive phosphorus is dissolved from rocks and becomes part of the sugar molecules created during photosynthesis.

(D) Radioactive phosphorus is dissolved from rocks and is used to create cholesterol stored in the nucleus.

4. 【U1 Chemistry of Life】U1 Q65 = U2 Q90

RADIOACTIVE COUNTS (PROTEIN X)

Sampling Time (minutes)	Cell Component						
	Endoplasmic Reticulum	Golgi Apparatus	Cytoplasm	Nucleus	Mitochondria	Lysosomes	Extracellular Space
5	980	20	50	4	2	6	2
20	135	760	20	3	5	120	9
45	50	195	4	6	2	800	3
120	15	30	8	6	8	975	4

65. What is the most likely identity of protein X ?

(A) An enzyme that is involved in transcription

(B) An electron transport protein that participates in respiration reactions

(C) A protein secreted from liver cells into the blood stream that permits proper capillary function

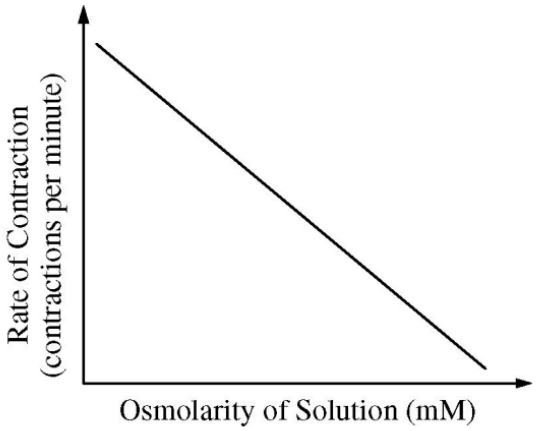
(D) An enzyme that participates in the degradation and recycling of cell components ✓

(E) A building block of the microtubules that are part of the cytoskeleton

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5. 【U2 Cell】U2 Q24 = U2 Q67

24.



Paramecia are unicellular protists that have contractile vacuoles to remove excess intracellular water. In an experimental investigation, paramecia were placed in salt solutions of increasing osmolarity. The rate at which the contractile vacuole contracted to pump out excess water was determined and plotted against osmolarity of the solutions, as shown in the graph. Which of the following is the correct explanation for the data?

(A) At higher osmolarity, lower rates of contraction are required because more salt diffuses into the paramecia.

(B) The contraction rate increases as the osmolarity decreases because the amount of water entering the paramecia by osmosis increases. ✓

(C) The contractile vacuole is less efficient in solutions of high osmolarity because of the reduced amount of ATP produced from cellular respiration.

(D) In an isosmotic salt solution, there is no diffusion of water into or out of the paramecia, so the contraction rate is zero.

6. 【U2 Cell】U2 Q25 = U2 Q26

25. A human kidney filters about 200 liters of blood each day. Approximately two liters of liquid and nutrient waste are excreted as urine. The remaining fluid and dissolved substances are reabsorbed and continue to circulate throughout the body. Antidiuretic hormone (ADH) is secreted in response to reduced plasma volume. ADH targets the collecting ducts in the kidney, stimulating the insertion of aquaporins into their plasma membranes and an increased reabsorption of water.

If ADH secretion is inhibited, which of the following would initially result?

(A) The number of aquaporins would increase in response to the inhibition of ADH.

(B) The person would decrease oral water intake to compensate for the inhibition of ADH.

(C) Blood filtration would increase to compensate for the lack of aquaporins.

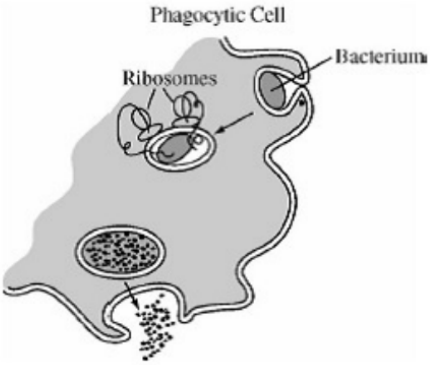
(D) The person would produce greater amounts of dilute urine. ✓

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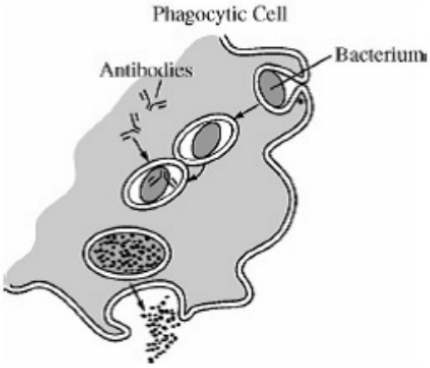
7. 【U2 Cell】U2 Q30 = U2 Q31

30. A pathogenic bacterium has been engulfed by a phagocytic cell as part of the nonspecific (innate) immune response. Which of the following illustrations best represents the response?

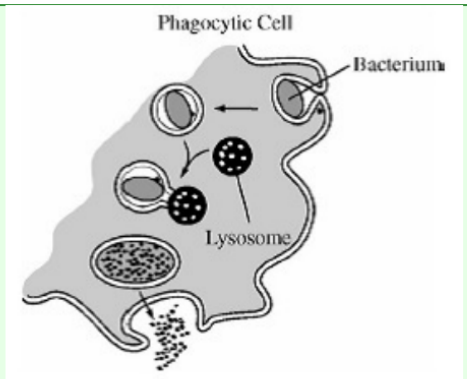
(A)


Phagocytic Cell
Bacterium
Ribosomes

(B)

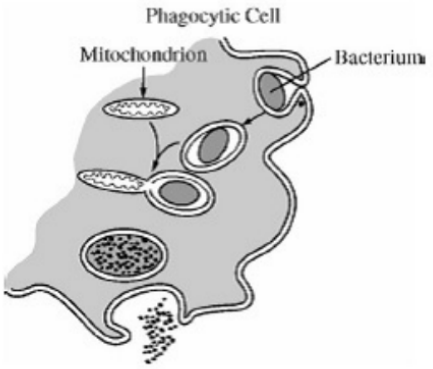

Phagocytic Cell
Bacterium
Antibodies

(C)


Phagocytic Cell
Bacterium
Lysosome

✓

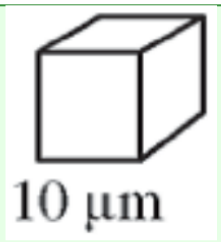
(D)

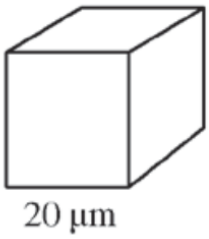


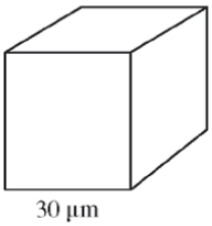
8. 【U2 Cell】U2 Q137 = U2 Q143

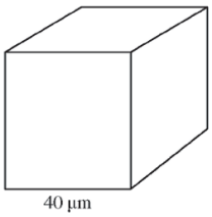
137. Simple cuboidal epithelial cells line the ducts of certain human exocrine glands. Various materials are transported into or out of the cells by diffusion. (The formula for the surface area of a cube is 6 X S^2, and the formula for the volume of a cube is S^3, where S = the length of a side of the cube.)

Which of the following cube-shaped cells would be most efficient in removing waste by diffusion?

(A) 

(B) 

(C) 

(D) 

Answer A

This option is correct. It demonstrates the ability to calculate surface area-to- volume ratios and predict their effects on a cell's ability to exchange materials with its environment. The cell shown in A has the greatest surface area-to- volume ratio at 0.6:1.

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9. 【U3 Cellular Energetics】U3 Q5 = U3 Q7

5. During aerobic cellular respiration, oxygen gas is consumed at the same rate as carbon dioxide gas is produced. In order to provide accurate volumetric measurements of oxygen gas consumption, the experimental setup should include which of the following?

(A) A substance that removes carbon dioxide gas ✓

(B) A plant to produce oxygen

(C) A glucose reserve

(D) A valve to release excess water

An experiment to measure the rate of respiration in crickets and mice at 10°C and 25°C was performed using a respirometer, an apparatus that measures changes in gas volume. Respiration was measured in mL of O_2 consumed per gram of organism over several five-minute trials and the following data were obtained.

10. 【U3 Cellular Energetics】U3 Q8 = U3 Q10

	Dish A		Dish B	
	Day 7 Covered	Day 14 Uncovered	Day 7 Uncovered	Day 14 Uncovered
Germinated seeds	12	20	20	20
Green-leaved seedlings	0	14	15	15
Yellow-leaved seedlings	12	6	5	5
Mean stem length below first set of leaves	8 mm	9 mm	3 mm	3 mm

8. According to the results of this experiment, germination of tobacco seeds during the first week is

(A) increased by exposure to light ✓

(B) unaffected by light intensity

(C) prevented by paper towels

(D) accelerated in green-leaved seedlings

Answer A

This option is correct. Dish B, which was uncovered, had 20 seeds germinate, while the covered dish A had only 12 seeds germinate in the first week.

11. 【U3 Cellular Energetics】U3 Q9 = U3 Q11

9. Additional observations were made on day 21, and no yellow-leaved seedlings were found alive in either dish. This is most likely because

(A) yellow-leaved seedlings were unable to absorb water from the paper towels

(B) taller green-leaved seedlings blocked the light and prevented photosynthesis

(C) yellow-leaved seedlings were unable to convert light energy to chemical energy ✓

(D) a higher rate of respiration in yellow-leaved seedlings depleted their stored nutrients

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Answer C

This option is correct. The only difference between the two plants after day 14 was the leaf color. Without chlorophyll, the yellow seedlings could not absorb enough light energy to convert to chemical energy to sustain growth. Therefore, no yellow seedlings were found alive in either dish after day 21.

A student placed 20 tobacco seeds of the same species on moist paper towels in each of two petri dishes. Dish A was wrapped completely in an opaque cover to exclude all light. Dish B was not wrapped. The dishes were placed equidistant from a light source set to a cycle of 14 hours of light and 10 hours of dark. All other conditions were the same for both dishes. The dishes were examined after 7 days, and the opaque cover was permanently removed from dish A. Both dishes were returned to the light and examined again at 14 days. The following data were obtained.

12. 【U3 Cellular Energetics】U3 Q47 = U3 Q48

47. Experimental evidence shows that the process of glycolysis is present and virtually identical in organisms from all three domains, Archaea, Bacteria, and Eukarya. Which of the following hypotheses could be best supported by this evidence?

(A) All organisms carry out glycolysis in mitochondria.

(B) Glycolysis is a universal energy-releasing process and therefore suggests a common ancestor for all forms of life. ✓

(C) Across the three domains, all organisms depend solely on the process of anaerobic respiration for ATP production.

(D) The presence of glycolysis as an energy-releasing process in all organisms suggests that convergent evolution occurred.

13. 【U4 Cell Communication and Cell Cycle】U4 Q3 = U5 Q93

3. A human cell in prophase of mitosis can be distinguished from a human cell in prophase I of meiosis by the presence of

(A) kinetochores in the mitotic cell

(B) a spindle in the mitotic cell

(C) twice as many chromosomes in the meiotic cell

(D) half as many chromosomes in the meiotic cell

(E) paired homologous chromosomes in the meiotic cell ✓

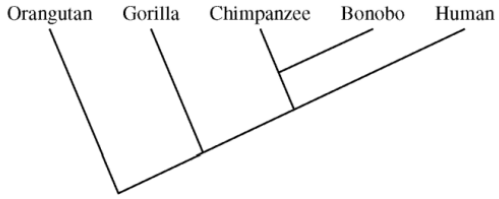
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14. 【U4 Cell Communication and Cell Cycle】U4 Q37 = U6 Q34

NUCLEOTIDE VARIATION IN THE *TAS2R38* GENE

	Position in the Nucleotide Sequence		
	145	785	886
Human nontaster	G	T	A
Human taster	C	C	G
Bonobo	C	C	G
Chimpanzee	C	C	G
Gorilla	C	C	G

A cladogram representing the evolutionary relatedness of selected primates is shown below.



37. The TAS2R38 receptor protein has been detected on the surface of cells from individuals who are homozygous for the nontaster allele of the *TAS2R38* gene. Which of the following is the most likely effect of the mutations associated with the nontaster allele on *TAS2R38* gene expression?

(A) The mutations change the primary structure of the encoded receptor protein. ✓

(B) The mutations increase the stability of the *TAS2R38* mRNA.

(C) The mutations prevent transcription of the *TAS2R38* gene.

(D) The mutations prevent translation of the *TAS2R38* mRNA.

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15. 【U4 Cell Communication and Cell Cycle】U4 Q49 = U4 Q50

49. The endocrine system incorporates feedback mechanisms that maintain homeostasis. Which of the following demonstrates negative feedback by the endocrine system?

(A) During labor, the fetus exerts pressure on the uterine wall, inducing the production of oxytocin, which stimulates uterine wall contraction. The contractions cause the fetus to further push on the wall, increasing the production of oxytocin.

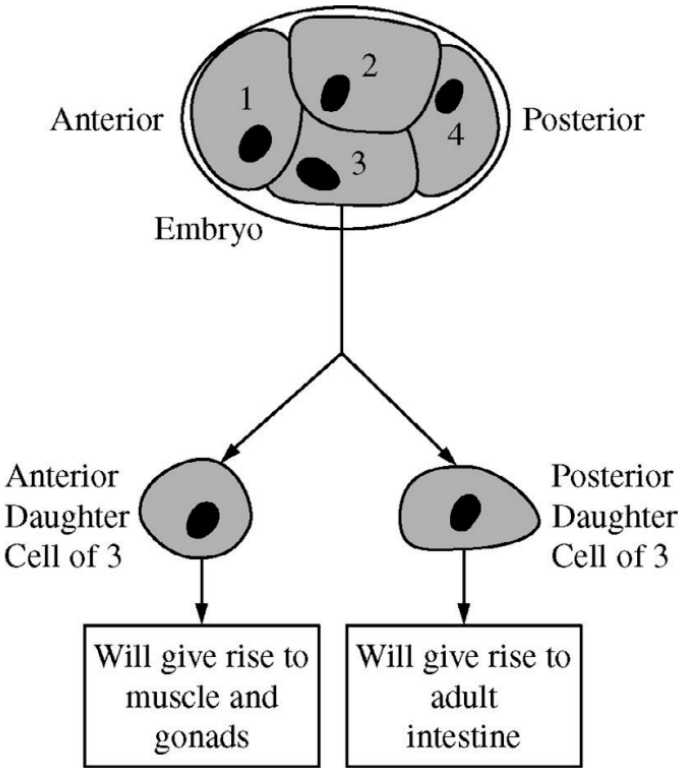
(B) After a meal, blood glucose levels become elevated, stimulating beta cells of the pancreas to release insulin into the blood. Excess glucose is then converted to glycogen in the liver, reducing blood glucose levels. ✓

(C) At high elevation, atmospheric oxygen is more scarce. In response to signals that oxygen is low, the brain decreases an individual's rate of respiration to compensate for the difference.

(D) A transcription factor binds to the regulatory region of a gene, blocking the binding of another transcription factor required for expression.

16. 【U4 Cell Communication and Cell Cycle】U4 Q99 = U4 Q100

99.



The diagram above shows a developing worm embryo at the four-cell stage. Experiments have shown that when cell 3 divides, the anterior daughter cell gives rise to muscle and gonads and the posterior daughter cell gives rise to the intestine. However, if the cells of the embryo are separated from one another early during the four-cell stage, no intestine will form. Other experiments have shown that if cell 3 and cell 4 are recombined after the initial separation, the posterior daughter cell of cell 3 will once again give rise to normal intestine. Which of the following is the most plausible explanation for these findings?

(A) A cell surface protein on cell 4 signals cell 3 to induce formation of the worm's intestine. ✓

(B) The plasma membrane of cell 4 interacts with the plasma membrane of the posterior portion of cell 3, causing invaginations that become microvilli.

(C) Cell 3 passes an electrical signal to cell 4, which induces differentiation in cell 4.

(D) Cell 4 transfers genetic material to cell 3, which directs the development of intestinal cells.

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17. 【U5 Heredity】U5 Q7 = U6 Q3

7. A genetic counselor is consulted by a young man who is worried about developing Huntington's disease, an inherited disorder caused by a dominant allele of a single gene. The young man explains that his cousin was recently diagnosed with Huntington's disease, and the news has caused him to consider his own risk of developing the disorder. Which of the following questions will best help the genetic counselor to evaluate the risk of the young man developing Huntington's disease and transmitting it to his children?

(A) Were you and your cousin born in the same geographical area?

(B) Were your parents or grandparents ever diagnosed with Huntington's disease? ✓

(C) Were you in physical contact with a person diagnosed with Huntington's disease?

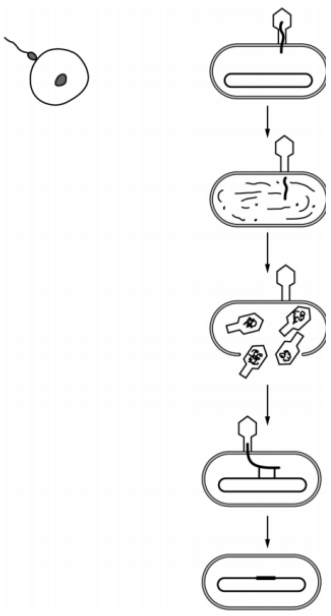
(D) Were you ever exposed to substances that are suspected of being mutagens?

Directions: Each group of questions below concerns an experimental or a laboratory situation. In each case, first study the description of the situation. Then choose the one best answer to each question following it.

In dogs, one pair of alleles determines coat color (dark and albino). Another pair of alleles determines hair length (short and long). Thus, each gamete will contain one of the coat-color alleles, *C* or *c* and one of the hair-length alleles, *B* or *b*. In repeated crosses of a specific dark, short-haired dog with an albino, long-haired dog, all the offspring were dark with short hair, as shown in cross I. However, in subsequent crosses of another dark, short-haired dog with a dark, long-haired dog, the ratios shown in cross II below were obtained.

18. 【U5 Heredity】U5 Q45 = U6 Q53

45.



Which of the following correctly compares the two processes shown above?

(A) Both processes increase genetic variation. ✓

(B) Both processes represent aspects of sexual reproduction.

(C) Both processes require partitioning of genetic material and organelles.

(D) The amount of genetic material per cell remains constant in both processes.

A student placed 20 tobacco seeds of the same species on moist paper towels in each of two petri dishes. Dish A was wrapped completely in an opaque cover to exclude all light. Dish B was not wrapped. The dishes were placed equidistant from a light source set to a cycle of 14 hours of light and 10 hours of dark. All other conditions were the same for both dishes. The dishes were examined after 7 days, and the opaque cover was permanently removed from dish A. Both dishes were returned to the light and examined again at 14 days. The following data were obtained.

	Dish A		Dish B	
	Day 7 Covered	Day 14 Uncovered	Day 7 Uncovered	Day 14 Uncovered
Germinated seeds	12	20	20	20
Green-leaved seedlings	0	14	15	15
Yellow-leaved seedlings	12	6	5	5
Mean stem length below first set of leaves	8 mm	9 mm	3 mm	3 mm

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19. 【U5 Heredity】U5 Q46 = U5 Q47

46. Which of the following best supports the hypothesis that the difference in leaf color is genetically controlled?

(A) The number of yellow-leaved seedlings in dish A on day 7

(B) The number of germinated seeds in dish A on days 7 and 14

(C) The death of all the yellow-leaved seedlings

(D) The existence of yellow-leaved seedlings as well as green-leaved ones on day 14 in dish B ✓

A student placed 20 tobacco seeds of the same species on moist paper towels in each of two petri dishes. Dish A was wrapped completely in an opaque cover to exclude all light. Dish B was not wrapped. The dishes were placed equidistant from a light source set to a cycle of 14 hours of light and 10 hours of dark. All other conditions were the same for both dishes. The dishes were examined after 7 days and the opaque cover was permanently removed from dish A. Both dishes were returned to the light and examined again at 14 days. The following data were obtained.

20. 【U5 Heredity】U5 Q86 = U5 Q91

86. A culture of white-eyed fruit flies (*Drosophila melanogaster*) was maintained for many generations. Females from the stock white-eyed culture were crossed with red-eyed (wild-type) males. The F_1 females were crossed with the white-eyed males from the original culture. The resulting phenotypes of the progeny are summarized below.

Parental Generation Cross	**F_1 Generation (at least 500 flies)**
White-eyed females × red-eyed males	100% of females are red-eyed 100% of males are white-eyed

F_1 Generation Cross	**F_2 Generation (at least 500 flies)**
F_1 red-eyed females × white-eyed males	50% of females are red-eyed and 50% are white-eyed 50% of males are red-eyed and 50% are white-eyed

There are white-eyed females in the F_2 generation because

(A) white is a dominant allele

(B) the white allele is autosomal

(C) a mutation has occurred

(D) these F_2 females have two white alleles ✓

(E) the white allele is located on the *Y* chromosome

21. 【U5 Heredity】U5 Q87 = U5 Q88

87. The tiny blue-eyed Mary flower is often one of the first flowers seen in the spring in some regions of the United States. The flower is normally blue, but sometimes a white or pink flower variation is found.

The following data were obtained after several crosses.

Parents	F_1	F_2
Blue x white	Blue	196 blue, 63 white
Blue x pink	Blue	149 blue, 52 pink
Pink x white	Blue	226 blue, 98 white, 77 pink

Which of the following statements best explains the data?

(A) The appearance of blue in the F_1 generation of the pink and white cross demonstrates that flower color is not an inherited trait but is determined by the environment.

(B) Flower color depends on stages of flower development, and young flowers are white, advancing to pink and then blue.

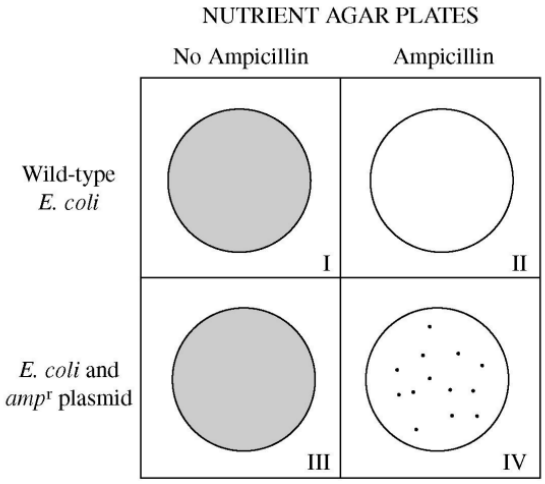
(C) Since the F_1 and F_2 phenotypes of the pink and white cross do not fit the expected genotypic and phenotypic ratios, blue-eyed Mary must reproduce by vegetative propagation.

(D) Flower color is an inherited trait, and the F_1 and F_2 phenotypes of the flowers arising from the pink and white cross can best be explained by another gene product that influences the phenotypic expression. ✓

Answer D

This option is correct. The student is asked to evaluate the data given. The crosses between blue / white and blue / pink show expected 3:1 ratios in the F_1 generation. The appearance of a majority of blue flowers in the F_1 cross between pink and white indicate that there is another gene product affecting the outcome.

22. 【U6 Gene Expression and Regulation】U6 Q4 = U6 Q9



4. Plates that have only ampicillin-resistant bacteria growing include which of the following?

(A) I only

(B) III only

(C) IV only ✓

(D) I and II

23. 【U6 Gene Expression and Regulation】U6 Q5 = U6 Q10

5. Which of the following best explains why there is no growth on plate II?

(A) The initial *E. coli* culture was not ampicillin-resistant. ✓

(B) The transformation procedure killed the bacteria.

(C) Nutrient agar inhibits *E. coli* growth.

(D) The bacteria on the plate were transformed.

24. 【U6 Gene Expression and Regulation】U6 Q6 = U6 Q11

6. Plates I and III were included in the experimental design in order to

(A) demonstrate that the *E. coli* cultures were viable ✓

(B) demonstrate that the plasmid can lose its amp^r gene

(C) demonstrate that the plasmid is needed for *E. coli* growth

(D) prepare the *E. coli* for transformation

25. 【U6 Gene Expression and Regulation】U6 Q7 = U6 Q12

7. Which of the following statements best explains why there are fewer colonies on plate IV than on plate III?

(A) Plate IV is the positive control.

(B) Not all *E. coli* cells are successfully transformed. ✓

(C) The bacteria on plate III did not mutate.

(D) The plasmid inhibits *E. coli* growth.

26. 【U6 Gene Expression and Regulation】U6 Q8 = U6 Q13

8. In a second experiment, the plasmid contained the gene for human insulin as well as the *amp*^{r} gene. Which of the following plates would have the highest percentage of bacteria that are expected to produce insulin?

(A) I only

(B) III only

(C) IV only ✓

(D) I and III

In a transformation experiment, a sample of *E. coli* bacteria was mixed with a plasmid containing the gene for resistance to the antibiotic ampicillin (*amp*^{r}). Plasmid was not added to a second sample. Samples were plated on nutrient agar plates, some of which were supplemented with the antibiotic ampicillin. The results of *E. coli* growth are summarized below. The shaded area represents extensive growth of bacteria; dots represent individual colonies of bacteria.

27. 【U6 Gene Expression and Regulation】U6 Q82 = U6 Q83

82. Sickle-cell anemia results from a point mutation in the *HBB* gene. The mutation results in the replacement of an amino acid that has a hydrophilic R-group with an amino acid that has a hydrophobic R-group on the exterior of the hemoglobin protein. Such a mutation would most likely result in altered

(A) properties of the molecule as a result of abnormal interactions between adjacent hemoglobin molecules ✓

(B) DNA structure as a result of abnormal hydrogen bonding between nitrogenous bases

(C) fatty acid structure as a result of changes in ionic interactions between adjacent fatty acid chains

(D) protein secondary structure as a result of abnormal hydrophobic interactions between R-groups in the backbone of the protein

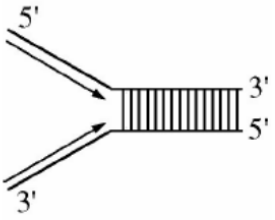
Answer A

This option is correct. The mutation results in the replacement of an amino acid, which alters the properties of the R-group both structurally and functionally; thus, the interactions between adjacent hemoglobin molecules would also be altered. In this case, hemoglobin molecules would tend to stick together as a result of hydrophobic interactions between amino acids on the surface.

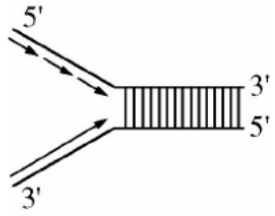
28. 【U6 Gene Expression and Regulation】U6 Q111 = U6 Q117

111. When DNA replicates, each strand of the original DNA molecule is used as a template for the synthesis of a second, complementary strand. Which of the following figures most accurately illustrates enzyme-mediated synthesis of new DNA at a replication fork?

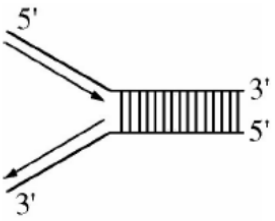
(A)



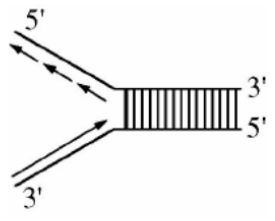
(B)



(C)



(D)



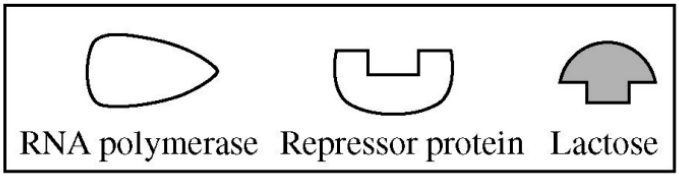
✓

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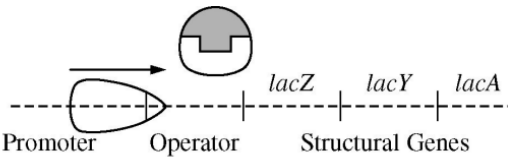
29. 【U6 Gene Expression and Regulation】U6 Q155 = U6 Q156

155. Lactose digestion in *E. coli* begins with its hydrolysis by the enzyme *β*-galactosidase. The gene encoding *β*-galactosidase, *lacZ*, is part of a coordinately regulated operon containing other genes required for lactose utilization.

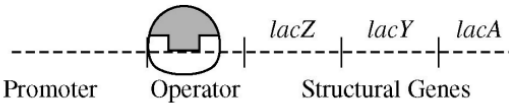
Which of the following figures correctly depicts the interactions at the *lac* operon when lactose is NOT being utilized? (The legend below defines the shapes of the molecules illustrated in the options.)



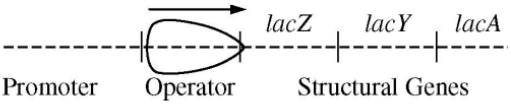
(A)



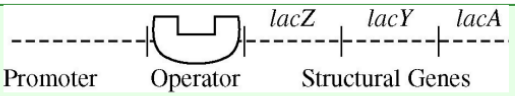
(B)



(C)



(D) ✓



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30. 【U7 Natural Selection】U7 Q22 = U7 Q23

22. By discharging electric sparks into a laboratory chamber atmosphere that consisted of water vapor, hydrogen gas, methane, and ammonia, Stanley Miller obtained data that showed that a number of organic molecules, including many amino acids, could be synthesized. Miller was attempting to model early Earth conditions as understood in the 1950s. The results of Miller's experiments best support which of the following hypotheses?

(A) The molecules essential to life today did not exist at the time Earth was first formed.

(B) The molecules essential to life today could not have been carried to the primordial Earth by a comet or meteorite.

(C) The molecules essential to life today could have formed under early Earth conditions. ✓

(D) The molecules essential to life today were initially self-replicating proteins that were synthesized approximately four billion years ago.

Answer C

This option is correct. It demonstrates the ability to evaluate scientific questions about the origin of life on Earth by recognizing that the Miller experiment modeled the presumed early atmospheric conditions and, under laboratory conditions, produced biological molecules, such as amino acids, that were required for early life.

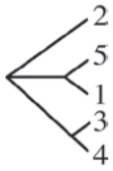
31. 【U7 Natural Selection】U7 Q52 = U7 Q53

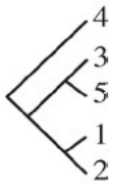
52. Five new species of bacteria were discovered in Antarctic ice core samples. The nucleotide (base) sequences of rRNA subunits were determined for the new species. The table below shows the number of nucleotide differences between the species.

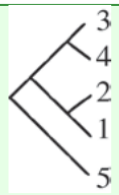
NUCLEOTIDE DIFFERENCES

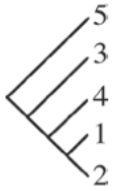
Species	1	2	3	4	5
1	-	3	19	18	27
2		-	19	18	26
3			-	1	27
4				-	27
5					-

Which of the following phylogenetic trees is most consistent with the data?

(A) 

(B) 

(C) 

(D) 

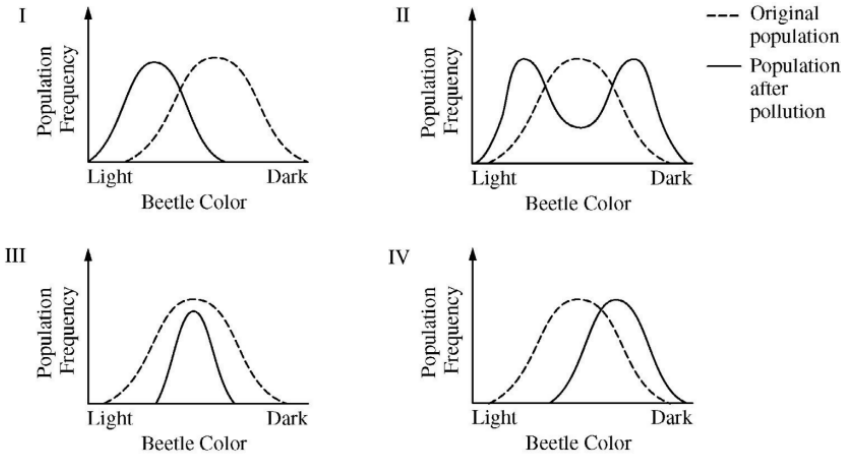
Answer C

This option is correct. It demonstrates the ability to select the phylogenetic tree that correctly represents evolutionary history and speciation from the data set. The data indicate that species 5 is not closely related to the others and that 3 and 4 are very closely related. This cladogram incorporates those differences as well as the close relationship between 1 and 2.

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32. 【U7 Natural Selection】U7 Q145 = U7 Q146

145. In a hypothetical population of beetles, there is a wide variety of color, matching the range of coloration of the tree trunks on which the beetles hide from predators. The graphs below illustrate four possible changes to the beetle population as a result of a change in the environment due to pollution that darkened the tree trunks.



Which of the following includes the most likely change in the coloration of the beetle population after pollution and a correct rationale for the change?

(A) The coloration range shifted toward more light-colored beetles, as in diagram I. The pollution helped the predators find the darkened tree trunks.

(B) The coloration in the population split into two extremes, as in diagram II. Both the lighter-colored and the darker-colored beetles were able to hide on the darker tree trunks.

(C) The coloration range became narrower, as in diagram III. The predators selected beetles at the color extremes.

(D) The coloration in the population shifted toward more darker-colored beetles, as in diagram IV. The lighter-colored beetles were found more easily by the predators than were the darker-colored beetles. ✓

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33. 【U8 Ecology】U8 Q4 = U8 Q5

4. A group of students summarized information on five great extinction events.

Mass Extinction	Time of Extinction	Organisms Greatly Reduced or Made Extinct
End of the Ordovician period	443 million years ago	Trilobites, brachiopods, echinoderms, and corals
End of the Devonian period	354 million years ago	Marine families on tropical reefs, corals, brachiopods, and bivalves
End of the Permian period	248 million years ago	Trilobites, mollusks, brachiopods, and many vertebrates
End of the Triassic period	206 million years ago	Mollusks, sponges, marine vertebrates, and large amphibians
End of the Cretaceous period	65 million years ago	Ammonites, dinosaurs, brachiopods, bivalves, and echinoderms

The students are sampling a site in search of fossils from the Devonian period. Based on the chart, which of the following would be the most reasonable plan for the students to follow?

(A) Searching horizontal rock layers in any class of rock and try to find those that contain the greatest number of fossils

(B) Collecting fossils from rock layers deposited prior to the Permian period that contain some early vertebrate bones ✓

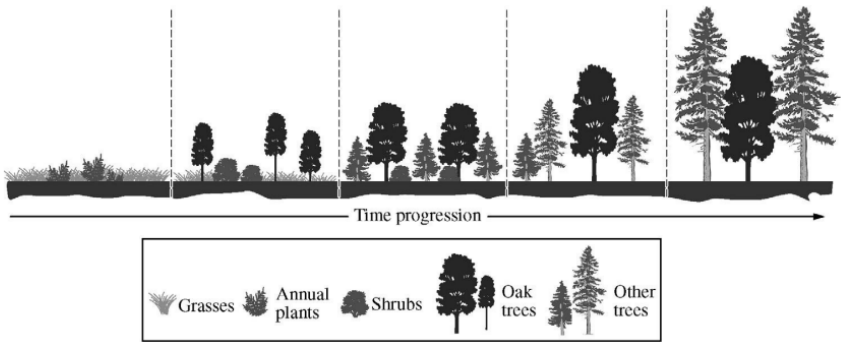
(C) Looking in sedimentary layers next to bodies of water in order to find marine fossils of bivalves and trilobites

(D) Using relative dating techniques to determine the geological ages of the fossils found so they can calculate the rate of speciation of early organisms

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34. 【U8 Ecology】U8 Q12 = U8 Q79

12.



The diagram above shows the progression of ecological events after a fire in a particular ecosystem. Based on the diagram, which of the following best explains why the oak trees are later replaced by other trees?

(A) Eventually the other trees grow taller than the oak trees and form a dense canopy that shades the understory. ✓

(B) Oak trees alter the pH of the soil, making the forest better suited for shrubs and other trees.

(C) Roots of shrubs proliferate in the soil of the forest and prevent the oak trees from obtaining water.

(D) Oak trees succumb to environmental pollutants more readily than do either the shrubs or the other trees.

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35. 【U8 Ecology】U8 Q91 = U8 Q92

91.

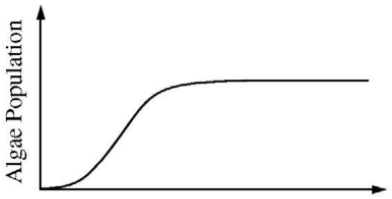


Figure I

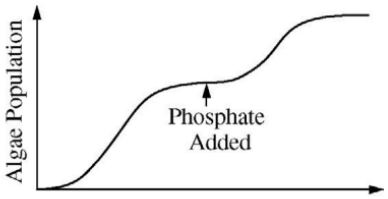
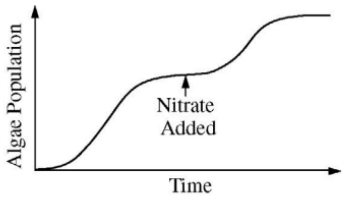


Figure II

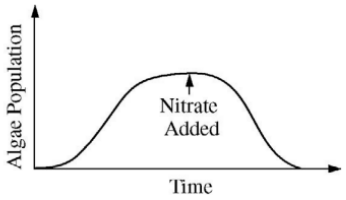
Figure I shows the growth of an algal species in a flask of sterilized pond water. If phosphate is added as indicated, the growth curve changes as shown in Figure II.

Which of the following is the best prediction of the algal growth if nitrate is added instead of phosphate?

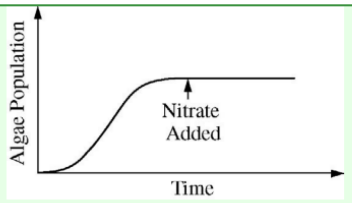
(A)



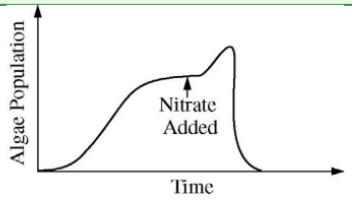
(B)



(C)



(D)



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36. 【U8 Ecology】U8 Q106 = U8 Q107

	Dish A		Dish B	
	Day 7 Covered	Day 14 Uncovered	Day 7 Uncovered	Day 14 Uncovered
Germinated seeds	12	20	20	20
Green-leaved seedlings	0	14	15	15
Yellow-leaved seedlings	12	6	5	5
Mean stem length below first set of leaves	8 mm	9 mm	3 mm	3 mm

106. The most probable cause for the difference in mean stem length between plants in dish A and plants in dish B is which of the following?

(A) Shortening of cells in the stem in response to the lack of light

(B) Elongation of seedlings in response to the lack of light ✓

(C) Enhancement of stem elongation by light

(D) Genetic differences between the seeds

Answer B

This option is correct. In response to the lack of light, the plant elongates cells in an attempt to reach light necessary for photosynthesis at the expense of leaf development.

A student placed 20 tobacco seeds of the same species on moist paper towels in each of two petri dishes. Dish A was wrapped completely in an opaque cover to exclude all light. Dish B was not wrapped. The dishes were placed equidistant from a light source set to a cycle of 14 hours of light and 10 hours of dark. All other conditions were the same for both dishes. The dishes were examined after 7 days, and the opaque cover was permanently removed from dish A. Both dishes were returned to the light and examined again at 14 days. The following data were obtained.

37. 【U8 Ecology】U8 Q130 = U8 Q131

130. Which of the following statements most directly supports the claim that different species of organisms use different metabolic strategies to meet their energy requirements for growth, reproduction, and homeostasis?

(A) During cold periods pond-dwelling animals can increase the number of unsaturated fatty acids in their cell membranes while some plants make antifreeze proteins to prevent ice crystal formation in tissues.

(B) Bacteria lack introns while many eukaryotic genes contain many of these intervening sequences.

(C) Carnivores have more teeth that are specialized for ripping food while herbivores have more teeth that are specialized for grinding food.

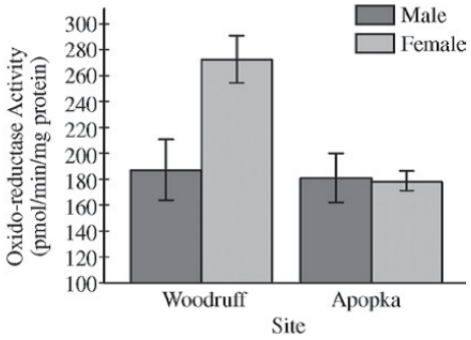
(D) Plants generally use starch molecules for storage while animals use glycogen and fats for storage. ✓

Answer D

This option is correct. Starch, glycogen, and fat are all molecules used for energy storage, yet plants have evolved to use starch while animals have evolved to use fat for long-term energy storage and glycogen for short-term energy storage.

38. 【U8 Ecology】U8 Q135 = U8 Q140

135.



Testosterone oxido-reductase is a liver enzyme that regulates testosterone levels in alligators. One study compared testosterone oxido-reductase activity between male and female alligators from Lake Woodruff, a relatively pristine environment, and from Lake Apopka, an area that has suffered severe contamination. The graph above depicts the findings of that study.

The data in the graph best support which of the following claims?

(A) Environmental contamination elevates total testosterone oxido-reductase activity in females.

(B) Environmental contamination reduces total testosterone oxido-reductase activity in females. ✓

(C) Environmental contamination elevates total testosterone oxido-reductase activity in males.

(D) Environmental contamination reduces total testosterone oxido-reductase activity in males.

Answer B

This option is correct. In Lake Woodruff, the lake that is relatively pristine, females had high levels of oxido-reductase enzyme activity, but in Lake Apopka, where environmental contamination conditions were severe, the enzyme activity was low, indicating that environmental factors had an influence on the organism.

Students investigated the effect of light on the carbon cycle in aquatic ecosystems by performing the controlled experiment summarized below. The students placed equal amounts of water (pH 7.0) from a large aquarium in glass beakers. The students transferred aquatic plants from the aquarium to several of the beakers, and then they placed equal numbers of the beakers in the light or the dark (Figure 1: groups I and II). Similarly, the students transferred goldfish from the same aquarium to other beakers, and then they placed equal numbers of those beakers in the light or dark (Figure 1: groups III and IV). Finally, the students placed an equal number of beakers containing water only in the light or dark (Figure 1: groups V and VI).

After exposing the samples to light or dark for one hour, the students recorded the pH of the water in each beaker. Carbon dioxide dissolved in water will lower the pH of an aqueous solution. In the experiment, the students used changes in pH to monitor changes in the amount of carbon dioxide in the water. For each treatment group, the students calculated the mean pH and standard error, as documented in the table below.

二、AP Biology 库内重复 —— FRQ 7 组（展示开头部分）

1. 【SG1 FRQ（=WB Phase One）】Q2 = Q5 = Q7

2. **Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.**

Hemoglobin is an iron-containing protein in red blood cells that transports oxygen from the lungs to cells throughout the body. Hemoglobin is a protein complex that is composed of four polypeptide subunits. Typically, two of the polypeptide subunits are encoded by the $HBA1$ gene and are called alpha-globin subunits, while the other two polypeptide subunits are encoded by the HBB gene and are called beta-globin subunits. Each globin subunit contains a single iron atom capable of binding oxygen.

Prior to birth and during early infancy, expression of the HBB gene is low and, instead, cells produce a third polypeptide called gamma-globin. During this time, the hemoglobin complex consists of two alpha-globin subunits and two gamma-globin subunits and is referred to as fetal hemoglobin. Fetal hemoglobin is able to bind oxygen more tightly than does the typical adult hemoglobin, which enables the fetus (unborn offspring) to remove oxygen from the hemoglobin in the mother's blood. A model of the expression levels of all three types of globin polypeptides during the weeks leading up to and following birth is shown in Figure 1.

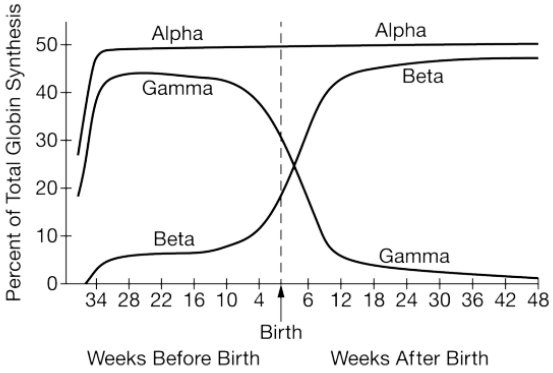
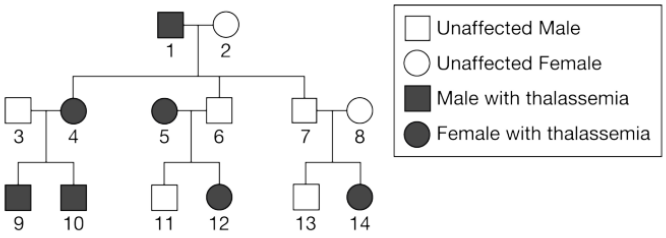


Figure 1. Model of globin expression in fetal and infant tissues

Mutations in the genes that encode globin subunits may affect the function of hemoglobin. Beta-thalassemia is an inherited disorder that results from a variety of mutations to the HBB gene that encodes the beta-globin subunit. The incidence of beta-thalassemia in one particular family is shown in Figure 2.



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2. 【SG1 FRQ（=WB Phase One）】Q8 = Q13

8. **Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.**

In plants, photosynthetic guard cells on the surface of leaves form pores called stomata (a single pore is called a stoma). Stomata help prevent water loss and control gas exchange with the environment to maintain plant homeostasis. The turgor pressure of the guard cells on either side of the stomata determines whether the stomata are open or closed based on the transport of certain ions into and out of the guard cells (Figure 1). A proton pump in the membrane of the guard cells pumps protons into the extracellular space. This makes the interior of the cell more negatively charged than the extracellular space.

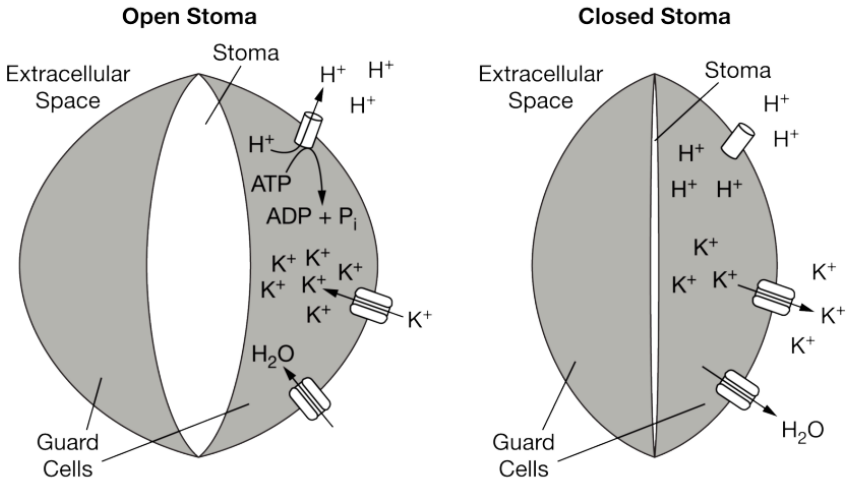


Figure 1. Models of open and closed stomata and the transport proteins in the guard cell membranes

Scientists interested in understanding drought tolerance in plants discovered a mutant variety of a plant with increased cell division, increased water content, and reduced density of leaf stomata compared with the wild type. Analysis of the mutant plant showed a mutation in the gene encoding a cyclin-dependent kinase (CDK). This mutation prevents CDK-cyclin interaction that normally results in the phosphorylation of RNA polymerase (Figure 2).

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3. 【SG1 FRQ（＝WB Phase One）】Q9 ＝ Q14

9. **Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.**

Aquaporin channel proteins are found in the plasma membrane of various cell types. Scientists studied the production of aquaporins in the keratinocytes (skin cells) of healthy individuals and found that these proteins play a role in hydration of the skin. Some individuals have a skin disorder that causes the absorption of excess water by the skin of their hands and feet when they are soaked in water. Scientists determined that the keratinocytes of these individuals are characterized by the presence of abnormal aquaporins. Scientists studied a family in which some individuals have this skin disorder and constructed a pedigree (Figure 1).

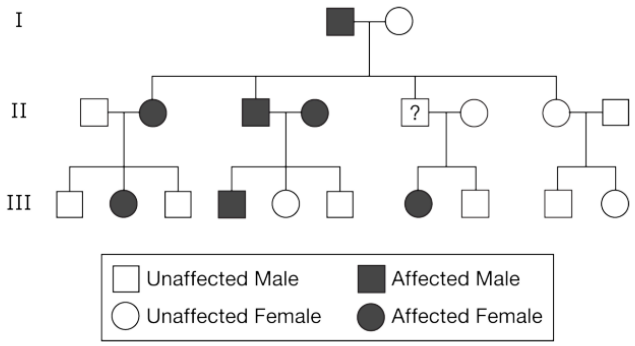


Figure 1. Pedigree of a human family in which some members have a skin disorder. An individual with an unknown phenotype is marked by a question mark.

(a) Based on the pedigree in Figure 1, **predict** the most likely genotype AND phenotype of the individual marked by "?". **Justify** your prediction.

(b) **Describe** the characteristics of the keratinocyte membrane that restrict water movement across the membrane. If isolated keratinocytes are placed in a hypotonic solution for 24 hours, **explain** why there are likely to be far fewer cells at the end of this time than there were at the start.

(c) **Explain** how the aquaporin polypeptides are transported from the Golgi complex and inserted into the plasma membrane.

(d) A student claims that individuals who have a lower than normal number of aquaporins in their skin cells will easily dehydrate. **Refute** this claim with evidence from biological concepts.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

4.【SG1 FRQ（＝WB Phase One）】Q10 ＝ Q12

10. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Increased expression and activity of the ligand-gated ion channel TRPV1 has been shown to block cell division and induce apoptosis in human cells. The structure of the TRPV1 protein consists of several transmembrane domains that are embedded in the membrane as well as a carboxy-terminus and an amino-terminus that are located inside the cell (Figure 1).

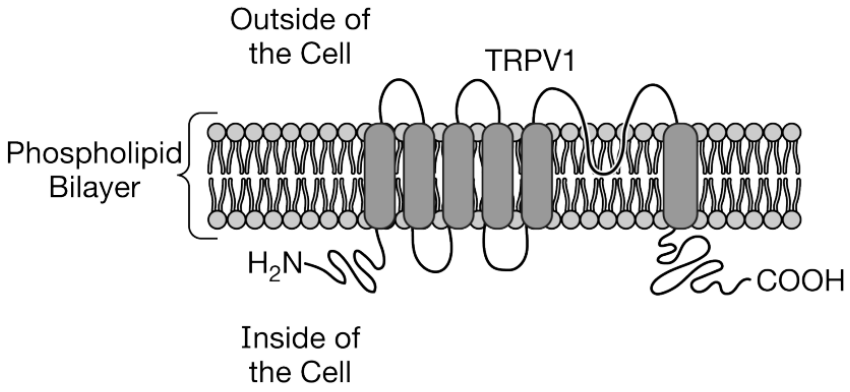


Figure 1. Structure of the TRPV1 protein channel in the membrane

TRPV1 is activated by several stimuli, including the binding of capsaicin, a chemical found in chili peppers. Capsaicin enters the cell by simple diffusion and then binds to one of the transmembrane domains of TRPV1, which opens the ion channel.

The opening of the TRPV1 channel allows Ca^{2+} ions to enter the cell, leading to the activation of the enzyme calcineurin. Calcineurin removes phosphate groups from the phosphorylated form of the transcription factor NFAT2, which is typically found in the cytoplasm. Once dephosphorylated, NFAT2 moves into the nucleus, where it blocks the transcription of another protein, ATF3. ATF3, when active, prevents the cell cycle regulatory protein pGoG from inducing apoptosis and blocking cell division (Figure 2).

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5. 【WB FRQ Phase Two】Q2 = Q4

2. **Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.**

In plants, a critical step in the pathway leading to the synthesis of the amino acid tryptophan includes the synthesis of an intermediate molecule, 5-enolpyruvylshikimic acid-3-phosphate (EPSP). The reactants in this step are shikimic acid-3-phosphate (S3P) and phosphoenolpyruvate (PEP), and the reaction is catalyzed by the enzyme EPSP synthase (Figure 1). Compound X, a chemical used to kill weeds, can affect EPSP synthesis by interacting with EPSP synthase.

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6. 【WB FRQ Phase Two】Q5 = Q11

5. **Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.**

Solid tumors are clusters of cancer cells and often contain blood vessels. When molecule B binds to the wild-type Brec protein in the plasma membrane of certain solid tumor cancer cells (Figure 1), the cancer cells express the membrane protein A and sometimes stimulate increased growth of blood vessels into the tumors.

Cells with a particular mutation in the *Brec* gene (Brec-MUT cells) have much increased expression levels of A and stimulate greater growth of blood vessels than do cancer cells with the wild-type Brec (Brec-WT cells); the cells with the mutant Brec can trigger intracellular signaling in the absence of B.

Researchers proposed that the signaling pathway modeled in Figure 1 is triggered by activation of the wild-type Brec and is associated with phosphorylation and activation of kinase D, expression of A, and the ability of the cancer cells to stimulate blood vessel growth.

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7. 【WB FRQ Phase Two】Q7 = Q9

7. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

The SIRT3 protein has been reported to increase the growth of certain types of cancer cells. Researchers studying the metabolism and development of stomach cancer cells investigated the potential role of the SIRT3 protein in these cells. In the first investigation, the researchers quantified the relative amount of SIRT3 in normal stomach cells and in each of four different lines of stomach cancer cells (Figure 1). Using data from a group of 100 patients with stomach cancer, the researchers also analyzed the relative amount of SIRT3 protein detected in samples of the cancer cells and in normal cells immediately adjacent to the cancer cells (Figure 2).

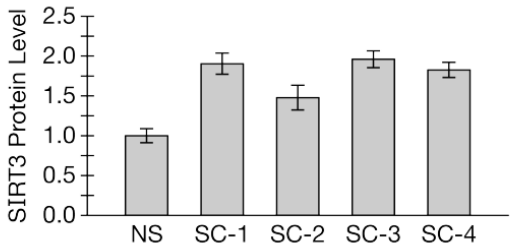


Figure 1. SIRT3 protein levels in normal stomach cells (NS) and stomach cancer cell lines from four different patients (SC-1, SC-2, SC-3, SC-4), shown relative to the level in normal stomach cells (NS). Error bars represent $\pm SE_{\bar{x}}$.

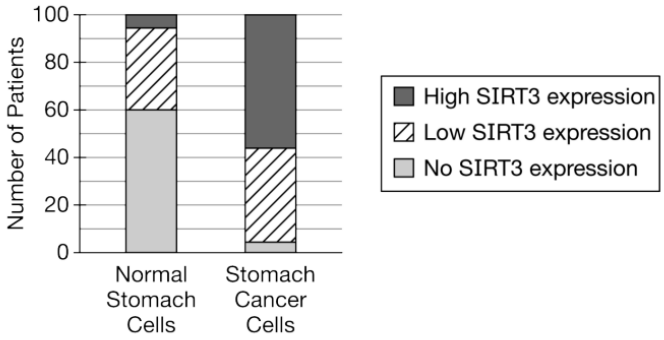


Figure 2. The number of patients, from 100 tested, whose stomach cancer cells or normal stomach cells adjacent to the cancer cells express high, low, or no SIRT3 protein

- In the second investigation, the researchers grew four groups of stomach cells in petri dishes containing nutrient broth:• NS, normal stomach cells• SC-1, a line of stomach cancer cells that the researchers had examined for SIRT3 expression• SC + RNA, the SC-1 line of stomach cancer cells to which a plasmid was added that encodes a small RNA complementary to a portion of the *SIRT3* mRNA. When these small RNAs bind to the complementary portion of *SIRT3* mRNA, the *SIRT3* mRNA is broken down rather than translated.• SC + plasmid, the SC-1 line of stomach cancer cells to which a plasmid was added that does NOT encode the complementary small RNA
- After 24 hours of growth, the researchers measured several indicators of metabolism by an equal number of all four cell groups: glucose uptake, lactic acid production, and the level of cytoplasmic ATP production in the presence of a compound that blocks the electron transport chain (Figure 3). The values were calculated relative to the values obtained for normal stomach cells.

三、AP Chemistry —— Progress Check MCQ 15 题全部与课时题库重复

1. Progress Check MCQ Q1 = 2.1 Q6

1. On the basis of electronegativity differences between atoms, which of the following scientific claims is the most accurate regarding the bonding in $Mg(NO_3)_2(s)$?

(A) There is polar covalent bonding between Mg atoms and N atoms.

(B) There is polar covalent bonding between Mg atoms and O atoms.

(C) There is ionic bonding between N^{5+} ions and O^{2-} ions.

(D) There is ionic bonding between Mg^{2+} ions and NO_3^- ions. ✓

Answer D

Correct. There is ionic bonding in solid $Mg(NO_3)_2$ due to coulombic attractions between Mg^{2+} ions and NO_3^- ions. (There is some polar covalent bonding between N and O within the NO_3^- ion.)

2. Progress Check MCQ Q2 = 2.1 Q14

2. Which of the following claims about a binary compound in which the bonding is ionic is most likely to be scientifically valid?

(A) Both elements in the compound are metals.

(B) The atomic masses of the elements in the compound are relatively small.

(C) There is equal sharing of electrons between the atoms of the elements in the compound.

(D) The electronegativity difference between the elements in the compound is relatively large. ✓

Answer D

Correct. A relatively large difference between the electronegativities of the two elements indicates that one element has a tendency to form cations and the other has a tendency to form anions.

3. Progress Check MCQ Q3 = 2.1 Q10

3. Which of the following claims about a binary compound composed of elements with the same electronegativity is most likely to be true?

(A) The compound has properties similar to those of both elements.

(B) The bonding in the compound is nonpolar covalent. ✓

(C) The boiling point of the compound is above $1000^\circ C$.

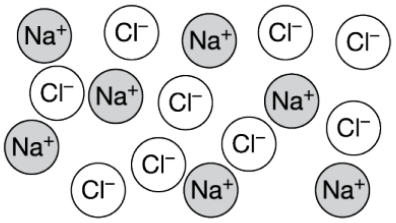
(D) The compound contains strong ionic bonds.

Answer B

Correct. Elements with the same electronegativity are more likely to share electrons and form a nonpolar covalent bond.

4. Progress Check MCQ Q4 = 2.3 Q5

4.



Molten (liquid) NaCl is represented by the particulate diagram shown above. Which of the following indicates whether $\mathrm{NaCl}(l)$ conducts electricity and best explains why or why not?

(A) It conducts electricity because Na is a metal.

(B) It conducts electricity because ions are free to move. ✓

(C) It does not conduct electricity because Cl is a nonmetal.

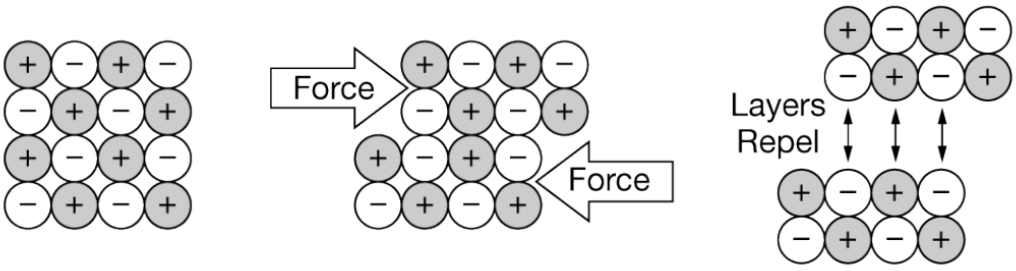
(D) It does not conduct electricity because there are no free electrons.

Answer B

Correct. For a substance to conduct electricity, it must contain charged particles that are mobile. In molten salts such as NaCl, those charged particles are ions. (Metals conduct electricity via mobile electrons within a lattice of metal atoms.)

5. Progress Check MCQ Q5 = 2.3 Q8

5.



The particulate-level diagram shown above best helps to explain which of the following properties of ionic solids?

(A) Density

(B) Brittleness ✓

(C) Malleability

(D) Conductivity

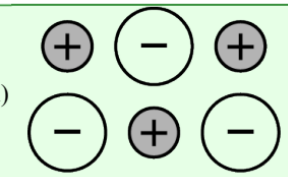
Answer B

Correct. The brittleness of ionic solids is due to the internal repulsion of ion layers within the crystal lattice that occurs when the layers are displaced by an outside force parallel to the layers and they move relative to adjacent layers.

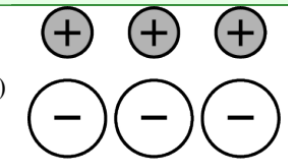
6. Progress Check MCQ Q6 = 2.3 Q3

6. The particles in solid KI, a stable ionic compound, are arranged to maximize coulombic attractions while minimizing coulombic repulsions among the particles. Which of the following diagrams best represents the structure of solid KI?

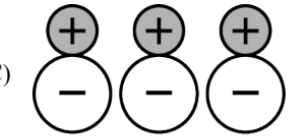
(A) ✓



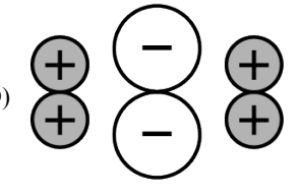
(B)



(C)



(D)

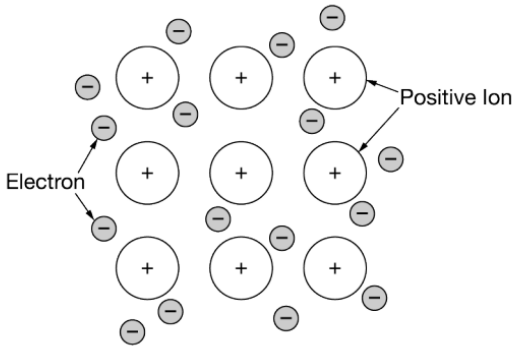


Answer A

Correct. This diagram shows both types of charged ions being surrounded by ions of the opposite charge, thus maximizing attraction and minimizing repulsion.

7. Progress Check MCQ Q7 = 2.4 Q3

7.



Which of the following indicates whether the solid substance represented by the particulate diagram shown above conducts electricity and explains why or why not?

(A) It conducts electricity because it is made of positive and negative particles.

(B) It conducts electricity because electrons are free to move through the substance. ✓

(C) It does not conduct electricity because electrons are strongly attracted to specific positive particles.

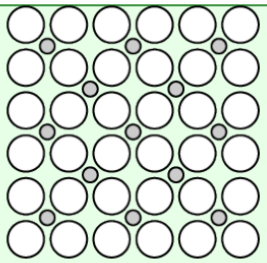
(D) It does not conduct electricity because the positive particles are not free to move through the substance.

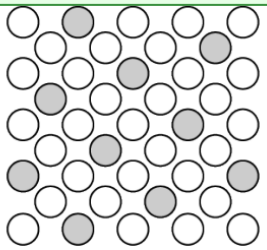
Answer B

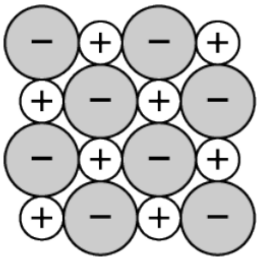
Correct. The diagram shows the structure of a metallic solid, which consists of a lattice of positive ions immersed in a sea of electrons in which electrons can move freely.

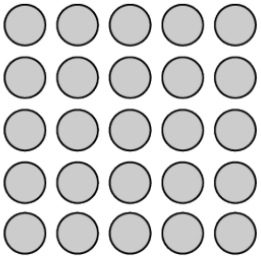
8. Progress Check MCQ Q8 = 2.4 Q5

8. Which of the following particulate-level diagrams best represents an interstitial alloy?

(A) 

(B) 

(C) 

(D) 

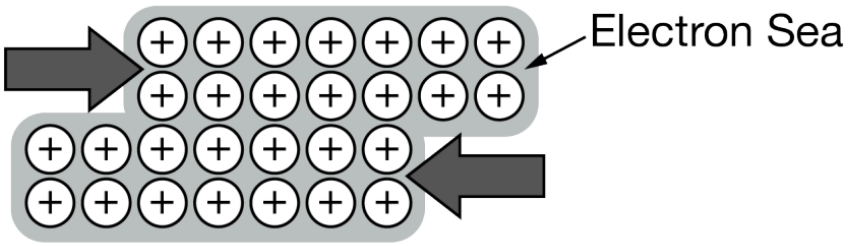
Answer A

Correct. In an interstitial alloy, atoms of one of the constituents are significantly smaller than the atoms of the other constituent and occupy the smaller spaces (interstices) among the larger atoms.

---

9. Progress Check MCQ Q9 = 2.4 Q9

9.



The diagram above best illustrates which of the following phenomena associated with solids that have metallic bonding?

(A) Electrical conductivity, because it shows a lattice of positive ions immersed in a sea of electrons.

(B) Malleability, because it shows how adjacent layers of positive ions can move relative to one another while remaining in full contact with the electron sea. ✓

(C) Heat conductivity, because it shows how layers of atoms can slide past one another, creating friction between layers and causing the temperature of the solid to increase.

(D) The ability to form substitutional alloys, because it shows how atoms of two different metals can combine in a one-to-one ratio.

Answer B

Correct. The diagram shows the structure of a metallic solid, which has adjacent layers of positive ions sliding past one another (as indicated by the arrows) while staying within the shading of the electron sea.

10. Progress Check MCQ Q10 = 2.6 Q6

10.

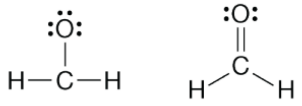


Diagram 1 Diagram 2

Which of the diagrams above best represents the CH_2O molecule, and why?

(A) Diagram 1, because all bond angles are 180°.

(B) Diagram 1, because all atoms have a formal charge of 0.

(C) Diagram 2, because the molecule has a trigonal pyramidal shape.

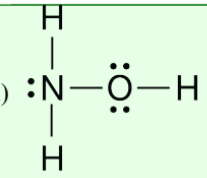
(D) Diagram 2, because all atoms have a formal charge of 0. ✓

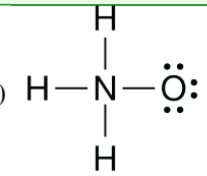
Answer D

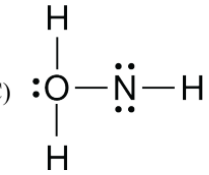
Correct. Diagram 2 represents a molecule in which both C and O have a complete octet and all atoms have a formal charge of zero.

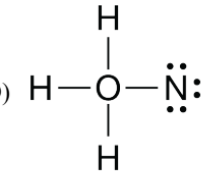
11. Progress Check MCQ Q11 = 2.6 Q8

11. Based on formal charges, which of the following is the best Lewis electron-dot diagram for H_3NO?

(A)  ✓

(B) 

(C) 

(D) 

Answer A

Correct. When the formal charge of each atom in this diagram is determined using the rule (formal charge of an atom = the number of valence electrons of the atom − (the number of electrons in lone pairs + half the number of electrons in bonds with other atoms)), the sum of the formal charges on all the atoms in H_3NO is equal to zero. A sum of formal charges on all atoms equal to zero is preferable.

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12. Progress Check MCQ Q12 = 2.6 Q7

12.



Diagram 1 Diagram 2

Which of the Lewis diagrams shown above is the more likely structure of CO_2, and why?

(A) Diagram 1, because all the atoms have a formal charge of 0. ✓

(B) Diagram 1, because double bonds are stronger than triple bonds.

(C) Diagram 2, because all the atoms have a formal charge of 0.

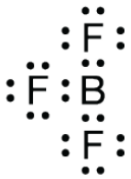
(D) Diagram 2, because triple bonds are stronger than double bonds.

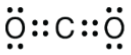
Answer A

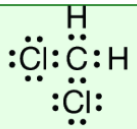
Correct. When the formal charge of each atom in diagram 1 is determined using the rule $($formal charge of an atom $=$ the number of valence electrons of the atom $-$ $($the number of electrons in lone pairs $+$ half the number of electrons in bonds with other atoms$))$, the results are zero for each of the three atoms. A molecular structure in which all formal charges are zero is preferable to one in which some formal charges are not zero. And in diagram 2, the results for the three atoms (going left to right) are -1, 0, and $+1$, respectively.

13. Progress Check MCQ Q13 = 2.7 Q40

13. Which of the following Lewis diagrams represents a molecule that is polar?

(A) 

(B) 

(C)  ✓

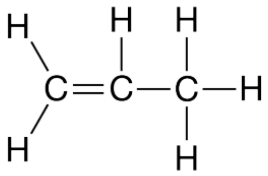
(D) 

Answer C

Correct. In the CH_2Cl_2 molecule, the H and Cl atoms are in a tetrahedral arrangement around the central C atom. The two electronegative chlorine atoms create a partial negative charge on their side of the carbon atom, leaving a partial positive charge on the opposite (hydrogen) side of the carbon atom. This separation of charge constitutes a dipole, and the molecule is polar.

14. Progress Check MCQ Q14 = 2.7 Q7

14.



Which of the following correctly compares the strength of the two carbon-to-carbon bonds in the molecule represented in the Lewis diagram shown above?

(A) The carbon-to-carbon bond on the left is stronger because it is a double bond. ✓

(B) The carbon-to-carbon bonds are the same strength because the C-C-C bond angle is 180°.

(C) The carbon-to-carbon bonds are the same strength because they are both bonds between carbon atoms.

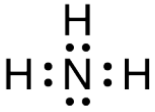
(D) The carbon-to-carbon bond on the right is stronger because there are more hydrogen atoms attached to the rightmost carbon atom than to the leftmost carbon atom.

Answer A

Correct. Double bonds between two atoms are always stronger than single bonds between those same two atoms because in a double bond there are more electrons between the two positive nuclei of the atoms, which results in a stronger bond.

15. Progress Check MCQ Q15 = 2.7 Q2

15.



Based on the Lewis diagram for NH_3, shown above, the H-N-H bond angle is closest to which of the following?

(A) 60°

(B) 90°

(C) 109.5° ✓

(D) 120°